Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Biodegradable Semiconductor–Based Electroceutical Platform for Targeted Spinal Cord Neuromodulation and Functional Recovery**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·Medical		*Electrical·Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Functional Restoration / Assistive & Welfare Devices	*Physical Function Restoration Devices*	*Semiconductor Devices and Systems*	*Semiconductor Devices for Medical Applications*
Concept of Proposed Technology	**Core Concept** The proposed technology is a biodegradable semiconductor–based electroceutical platform designed to promote neural regeneration and functional recovery after spinal cord injury through targeted neuromodulation (Figure 1). Unlike conventional implantable neuromodulation devices made of permanent metallic and polymeric materials, this platform integrates biodegradable electrodes, semiconductor components, and encapsulation layers that maintain stable electrical performance during the therapeutic window and subsequently undergo controlled bioresorption in vivo. This eliminates the need for secondary removal surgery and significantly reduces long-term complication risks. **Key Functions and Potential Applications**			

The system enables precise spatiotemporal electrical stimulation at the injured spinal cord segment, directly modulating neural circuits responsible for plasticity and regeneration rather than merely inducing peripheral muscle activation.

In addition, stimulation performance and biological responses are systematically evaluated using an AI-assisted testing and validation framework, allowing data-driven optimization of stimulation parameters based on functional and electrophysiological outcomes.

Potential applications include next-generation electroceutical therapies for spinal cord injury, implantable neuromodulation devices requiring temporary intervention, and future extensions to other central and peripheral nervous system disorders.

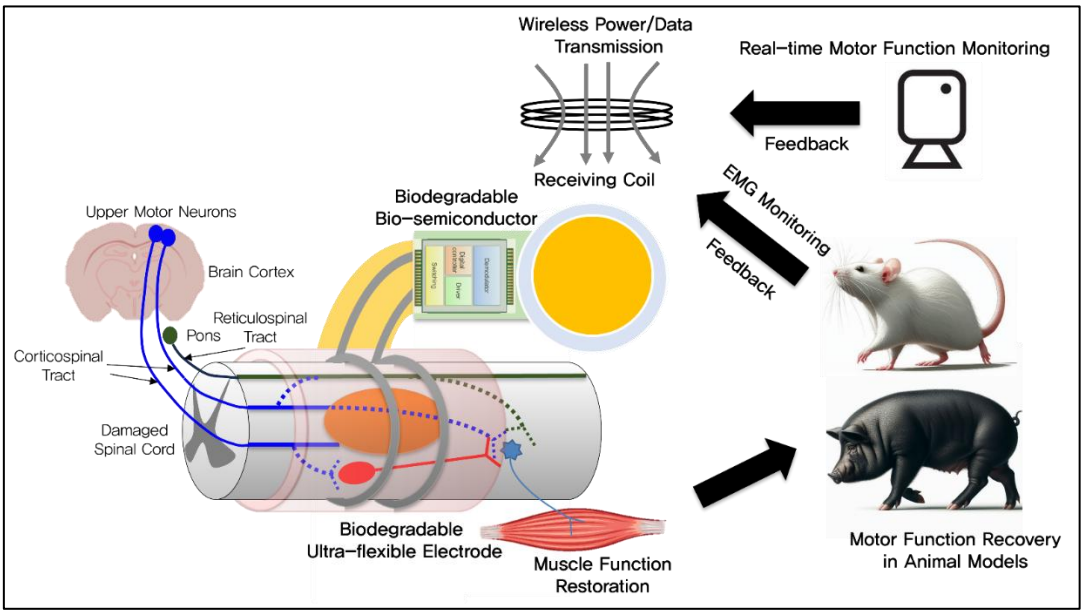

Figure 1. System Architecture and Functional Mechanism of the Biodegradable Electroceutical Platform

Figure 1 illustrates the comprehensive framework for targeted spinal cord neuromodulation using a biodegradable bio-semiconductor system. The platform establishes a closed-loop feedback mechanism where wireless power and data are transmitted to an implanted receiving coil, which then interfaces with a biodegradable bio-semiconductor and ultra-flexible electrodes. These electrodes are designed to conformally contact the injured spinal cord, modulating key neural pathways such as the corticospinal and reticulospinal tracts to promote plasticity. Real-time motor function and electromyography (EMG) monitoring provide continuous feedback, enabling AI-assisted optimization of stimulation parameters. The efficacy of the system in restoring muscle function and achieving behavioral recovery is validated through established animal models.

Rationale for Joint R&D Project

Critical Technical Barriers

The successful development of biodegradable electroceuticals requires tight integration of advanced semiconductor processing, bioresorbable materials engineering, system-level device design, and biological validation, which cannot be achieved by a single institution.

In particular, the core technical bottleneck lies in the transfer and integration of high-performance

	semiconductor devices onto biodegradable, ultra-flexible substrates without compromising electrical integrity. **Why Global Collaboration?** This capability relies on proprietary large-area transfer and system integration technologies that have been developed over more than a decade by the overseas research partner. Meanwhile, the Korean SME partner contributes stimulation chip development and an AI-assisted testing platform that enables systematic evaluation and optimization of neuromodulation parameters—bridging the critical gap between device-level performance and biological efficacy. **Expected Impact** Through this international joint R&D framework, the project aims to accelerate technology maturation from laboratory-scale validation to system-level prototype demonstration, reduce development risk, and establish a globally competitive electroceutical platform with clear commercialization potential.
Key Words	1. Biodegradable semiconductor 2. Electroceuticals 3. Spinal cord injury 4. Neuromodulation 5. AI-assisted stimulation optimization

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	**R&D Objectives** The primary objective of this joint R&D project is to develop and validate a system-integrated biodegradable electroceutical platform capable of delivering stable, targeted neuromodulation for functional recovery after spinal cord injury. **Detailed Objectives** 1) Development of biodegradable semiconductor-based stimulation modules ● Establish reliable fabrication and transfer processes for semiconductor devices integrated onto biodegradable, ultra-flexible substrates. 2) System-level integration of electroceutical components ● Integrate biodegradable electrodes, stimulation ICs, encapsulation layers, and wireless interfaces into a unified, implantable prototype. 3) Ensuring functional stability within the therapeutic window ● Achieve predictable electrical performance and mechanical integrity throughout the intended treatment duration, followed by controlled bioresorption. 4) AI-assisted stimulation testing and optimization ● Implement a data-driven testing framework to evaluate neuromodulation efficacy and optimize stimulation parameters based on electrophysiological and functional outcomes. 5) Preclinical validation of safety and functional efficacy ● Demonstrate system-level performance and biological relevance in established spinal cord injury models.			
R&D Target Specifications	The project aims to achieve the following target specifications at the system level: 1) Electrical performance ● Stable stimulation output within the therapeutic range required for spinal cord neuromodulation ● Electrode-tissue interface impedance maintained below predefined thresholds during the active treatment period ● Signal-to-noise ratio sufficient for reliable stimulation and monitoring in vivo 2) Mechanical and material properties ● Ultra-flexible device architecture enabling conformal contact with spinal tissue ● Mechanical durability under physiological deformation without fracture or delamination 3) Biodegradation characteristics			

	● Controlled bioresorption initiated after completion of the therapeutic window ● Absence of abrupt electrical failure prior to intended degradation ● Biocompatible degradation byproducts with no observable long-term tissue toxicity 4) System-level functionality ● Successful integration of stimulation, control, and encapsulation components into a single implantable prototype ● Compatibility with AI-assisted testing pipelines for stimulation parameter optimization 5) Preclinical performance ● Demonstrable modulation of neural activity associated with functional recovery endpoints in spinal cord injury models ● Quantitative thresholds and acceptance criteria will be finalized and refined during early-stage validation based on experimental data and regulatory guidance.
R&D Plan	**Year 1: Core Materials and Component Validation** ● Fabrication and characterization of biodegradable electrodes, semiconductor elements, and encapsulation layers ● Electrical, mechanical, and degradation profiling at the component level ● Selection of optimized material combinations for system integration ● **Milestone:** Validated biodegradable components meeting baseline electrical and mechanical criteria (TRL 4) **Year 2: Semiconductor Transfer and System Integration & AI-Assisted Stimulation Testing Platform** ● F Large-area transfer integration of semiconductor devices onto biodegradable substrates ● Assembly of stimulation ICs, electrodes, and encapsulation into a unified device architecture ● Bench-top electrical and mechanical reliability testing ● Development of an AI-assisted testing framework to analyze stimulation-response datasets ● Parameter space exploration for stimulation waveforms, amplitudes, and timing ● Feedback-driven refinement of stimulation protocols ● **Milestone:** Functionally integrated electroceutical prototype with stable in vitro performance (TRL 5) **Year 3: Preclinical System-Level Validation (Figure 2)** ● Implantation and evaluation of the integrated prototype in relevant spinal cord injury models ● Assessment of safety, electrical stability, biodegradation behavior, and functional outcomes ● Correlation of stimulation parameters with biological and behavioral endpoints ● **Milestone:** System-level demonstration in a relevant biological environment (TRL 6)

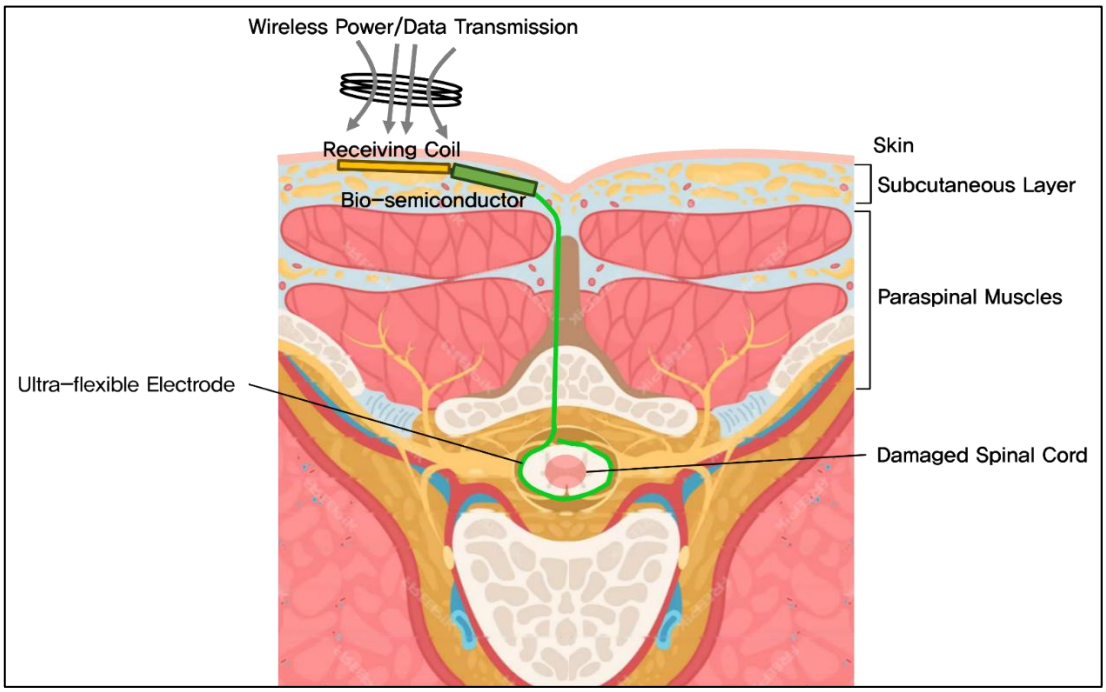

Figure 2. Anatomical Integration and Interface of the Implantable Biodegradable Device

Figure 2 provides a cross-sectional schematic demonstrating the anatomical placement of the electroceutical device within the spinal microenvironment. The power and data receiving coil is positioned within the subcutaneous tissue for efficient wireless coupling, while the bio-semiconductor module manages signal processing. The biodegradable, ultra-flexible electrode is strategically wrapped around the injured spinal cord segment, traversing the paraspinal muscles to deliver precise electrical stimulation. This integrated architecture ensures stable therapeutic intervention during the critical recovery window, followed by controlled bioresorption of all implanted components, thereby eliminating the need for secondary surgical removal and minimizing long-term tissue complications.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**The research institute(s), including Purdue University, will be responsible** The research institute(s), led by the overseas partner, will be responsible for core technology development and system-enabling research that cannot be readily replicated by industrial partners within the project timeframe. 1) Advanced semiconductor transfer and integration technologies ● Development and optimization of large-area transfer processes for integrating high-performance semiconductor devices onto biodegradable and ultra-flexible substrates ● Ensuring electrical integrity, alignment accuracy, and yield during transfer and system assembly 2) Materials and device-level characterization ● Electrical, mechanical, and degradation characterization of biodegradable semiconductor components and electrode systems ● Establishment of performance baselines and failure modes critical for system reliability 3) System-enabling design methodologies ● Contribution to device architecture design, including layout strategies for conformal neuromodulation at the spinal cord ● Technical guidance on scalability, reproducibility, and integration constraints 4) Research infrastructure and expertise ● Provision of specialized fabrication facilities, characterization tools, and expert personnel required for semiconductor processing and transfer ● Active participation in joint technical reviews and milestone evaluations
Expected R&R of Korean SME Partner	The Korean SME partner will take primary responsibility for device realization, system validation, and translational readiness, bridging research outcomes with practical electroceutical implementation. 1) Development of stimulation chips and control modules ● Design and fabrication of stimulation ICs optimized for biodegradable electroceutical systems ● Integration of stimulation control circuitry with biodegradable electrodes and encapsulation layers 2) AI-assisted testing and validation platform ● Development of an AI-assisted testing framework to analyze stimulation-response datasets ● Data-driven optimization of stimulation parameters, including waveform, amplitude, and temporal patterns ● Performance benchmarking linking device-level outputs to biological and functional outcomes 3) System integration and prototype realization

	● Assembly of system-level electroceutical prototypes incorporating stimulation chips, electrodes, and packaging ● Iterative refinement based on feedback from preclinical testing and joint technical reviews 4) Preclinical testing and translational support ● Execution of system-level validation in relevant experimental settings ● Documentation of performance, safety, and reliability data aligned with future regulatory and commercialization requirements 5) Collaboration and resource commitment ● Dedicated engineering and data analysis personnel assigned to the project ● Active coordination with the research institute to ensure seamless technology transfer and risk mitigation

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Outcomes 1) System-level biodegradable electroceutical platform (TRL 6) ● First-in-class demonstration of a system-integrated electroceutical device in which electrodes, semiconductor components, and encapsulation layers are all biodegradable, validated in a relevant biological environment. 2) Validated semiconductor transfer and integration methodology ● Establishment of reproducible processes for integrating high-performance semiconductor devices onto biodegradable, ultra-flexible substrates without loss of electrical integrity. 3) Data-driven neuromodulation optimization framework ● Implementation of an AI-assisted testing platform enabling systematic optimization of stimulation parameters, directly linking device-level outputs with electrophysiological and functional recovery endpoints. 4) Preclinical safety and efficacy dataset ● Comprehensive system-level data supporting electrical stability, controlled biodegradation, and functional relevance, forming a strong basis for subsequent regulatory and clinical development. Commercialization and Translational Impact 1) Elimination of secondary removal surgery ● By enabling complete bioresorption after the therapeutic window, the technology fundamentally addresses a major clinical and economic limitation of current implantable neuromodulation devices. 2) Accelerated pathway to next-stage product development ● The project outcomes will position the technology at a stage suitable for follow-on development toward clinical trials, regulatory engagement, and strategic partnerships with global medical device companies. 3) Platform scalability ● Although initially focused on spinal cord injury, the underlying platform is broadly applicable to other indications requiring temporary neuromodulation, including brain lesions, peripheral nerve injury and postoperative neural rehabilitation. Industrial and Economic Impact 1) Strengthening of national competitiveness in electroceuticals ● The project contributes to establishing a domestic technological base in biodegradable semiconductor–based medical devices, a field currently dominated by overseas players using permanent implant technologies. 2) High-value SME-driven innovation

	● By positioning the Korean SME partner as a system integrator and validation leader, the project supports sustainable SME growth and technology-driven job creation. 3) Catalyst for international collaboration and investment ● The validated technology platform and international joint R&D structure are expected to facilitate follow-on global collaborations, licensing opportunities, and private investment. Broader Societal Impact 1) Reduced patient burden and healthcare costs ● Minimization of surgical interventions and long-term implant-related complications directly benefits patient safety and healthcare system efficiency. 2) Advancement of precision neuromodulation paradigms ● The project advances the shift from empirical electrical stimulation toward mechanism-informed, data-driven neuromodulation, contributing to the long-term evolution of electroceutical medicine.

6. References

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